

switching times offer the advantage of using a single light modulator i.e., one DMD, to produce 256 shades per primary colour, as compared to other slower modulators which require three separate modulators for each primary colour.

Thus, using a single DMD, smaller and more compact projectors can be made, which offer results on par or even better than projectors based on slower light modulators, as in LCD technologies, for instance.

DLP In Projectors

Depending on the number of DMDs used (one, two, or three), three configurations of DLP projectors are currently available in the market. Remember, the choice of configuration depends upon the intended application, cost, brightness levels, lamp technology, weight and power dissipation.

One-chip configurations are normally used in consumer grade products such as conference room projectors and televisions. The single chip projector is cheaper, offers decent brightness levels and results in smaller, more portable designs.

Generally, the two chip projectors are used in situations where a longer lamp life is desired since they offer better light efficiency.

Three chip projectors offer the highest optical efficiency, and hence are used in



venues where large-screen projection is required, such as public information displays, auditoriums and movie theatres. Shown on the previous page is a functional schematic of a three chip DLP projector.

Apart from the above-mentioned parameters, 'Resolution' of the projected image is also important. Products using DLP technology capable of SVGA (800 x 600), XGA (1024 x 768) and SXGA (1280 x 1024) are available in the market. Recently, improvements have added high definition capabilities to DMDs, and such products will be available soon.

DLP Cinema

The goal of most multimedia projector manufacturers is to offer a product that can rival the quality of 35 mm film projection. The motion picture industry has long since shifted to digital technology, in terms of the production of movies; however, movie theatres still use last century's technology to show them to us.

In recent years, the motion picture industry has opted for multi-track sound formats such as Dolby or DTS, resulting in a dramatically improved aural experience. The visual experience though, remains outdated.

Moreover, film-based projection has severe limitations—duplication and distribution is expensive, deteriorating film quality with repeated screening, focus flutters, travel ghosting and inconvenient updating of post production content such as advertisements.

An exciting development in DLP technology is all set to change this with the introduction of DLP Cinema. The goal of DLP Cinema is to make the entire motion picture industry a digital industry, right from the production and distribution, to the final presentation of the film.

In 1997, a prototype using a 1280 x 1024 (SXGA) DMD called the DMD1210 was made, for evaluating the rapid upgrading of existing projectors to produce high quality, high brightness images. It involved the technical and creative community of the motion picture industry. The DLP Cinema technology demonstration projector exceeded the expectations of most people.

Although, the DLP Cinema has apparent advantages over film based projection, as was proved by the demonstration, there are some hurdles that the technology has to see through. A considerable amount of work is needed in terms of data compression, storage, and media delivery, before the 'Cinema of Tomorrow' can materialise. ☒

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DLP Versus LCD

LCD competes directly with DLP in the projector segment, and hence the question arises, "Which is better?" The fact of the matter is that neither of them has a clear advantage over the other. Let's have a quick look at the advantages and disadvantages of both.

Historically, LCD projectors have displayed better colour saturation than DLP projectors. This is because DLP projectors have a clear white panel in the colour wheel to boost brightness, due to which, the picture appears to be brighter than normal, and causes colours to look dull. However, the latest DLP projectors have a six-segment colour wheel, eliminating the white component. Despite this, LCD projectors still seem to offer better colour saturation.

On the other hand, DLP projectors offer better contrast video, with deeper black shades than what LCDs are capable of. Leading LCD projectors are rated to have 1000:1 contrast ratio, whereas a DLP projector is rated at 3000:1.

LCD projectors offer sharper pictures at any resolution, when compared to DLP projectors. The difference in sharpness is seen better when viewing high-detail images of financial sheets, rather than during the playback of video. Of course, this does not mean that DLP projected images are blotchy; the differences can only be made out when a DLP projection is compared side-by-side with an LCD projection.

Unlike LCD projectors, DLP projectors are not plagued by pixelation problems. Nevertheless, advances in LCD technology are overcoming this limitation, and given time, even cheaper LCD projectors will incorporate these developments.

The most obvious advantage of DLP is the compact size of units that use a single chip solution. Though LCD technology also aims at making smaller, more compact projectors, they have a long way to go to catch up with DLP solutions.

Market Overview

DLP projectors are manufactured by most consumer product manufacturers including Sony, Panasonic, Toshiba, NEC, Mitsubishi and InFocus. IT companies like BenQ, Acer and Dell have also entered this field and have at least one DLP model. The cheapest DLP projector is available for Rs 60,000, but the price can go as high as Rs 3 lakh for top-end models, which offer higher resolutions and better features.